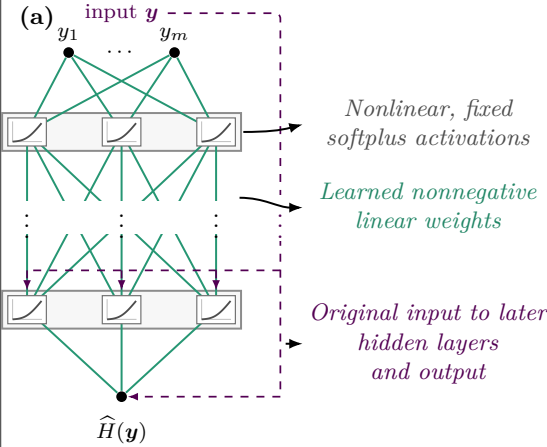
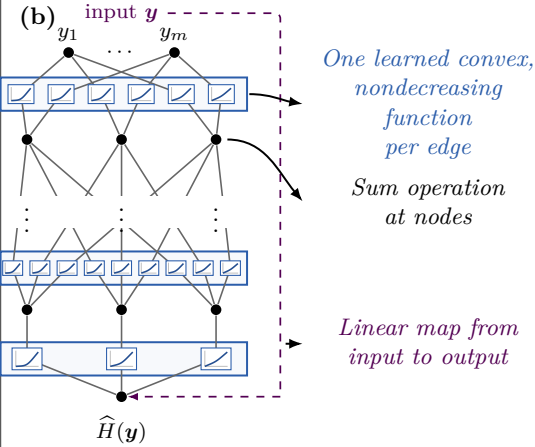


Core	Monotone ICNN	Monotone integrated-hat ICKAN
Layer operation	Learned nonnegative linear weights + biases $\rightarrow$ fixed softplus at hidden nodes	One learned scalar function per edge (convex and nondecreasing) $\rightarrow$ sum at nodes + biases
Multilayer core	 (a) input y $y_1 \dots y_m$ Nonlinear, fixed softplus activations Learned nonnegative linear weights Original input to later hidden layers and output $\hat{H}(y)$	 (b) input y $y_1 \dots y_m$ One learned convex, nondecreasing function per edge Sum operation at nodes Linear map from input to output $\hat{H}(y)$
	— Learned linear weights ▢ Fixed softplus	▢ Learned spline functions - - - Input maps